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Design, synthesis of pyridine coupled pyrimidinone/pyrimidinthione as anti-MRSA agent: Validation by molecular docking and dynamics simulation

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ABSTRACT

Methicillin Resistant *Staphylococcus aureus* (MRSA) is a major cause of severe hospital and infections acquired by the population and related morbidity and mortality. In this unique situation, there is a need of dynamic strong drug candidates to control MRSA diseases. Thus, the present work focuses on the synthesis and characterization of pyrimidinones and pyrimidinthiones coupled pyridine derivatives as anti-MRSA agent. The synthesized compounds were characterized by different spectroscopic techniques and evaluated against MRSA strain. Among them, **4e** and **4g** possessed better antibacterial activity with MIC values of 10 µg and 8 µg respectively. The key determinant of the wide range beta-lactam resistance in MRSA strains is the Penicillin-Binding Protein 2a (PBP2a) but the gene encodes PBP2a which has a low affinity towards β-lactam antibiotics. Because of this, the present investigation focused on the mechanism of PBP2a protein binding studies by *in-silico* studies. The synthesized compounds showed very good interactions with PBP2A compared with standard drug Vancomycin, among them compound **4g** showed better interaction with the binding score of −9.8 kcal/mol. Antibacterial activity was validated with molecular docking and molecular dynamic simulation. Simulation results revealed that protein-ligand interactions of **4g** compound stably sustained up to 20,000ps.

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KEYWORDS
MRSA; molecular docking; molecular dynamics; PBP2a; Pyridine

1. Introduction

The multi-drug resistance of *Staphylococcus aureus* has become a significant clinical issue around the world and the rise of Methicillin-resistant *Staphylococcus aureus* (MRSA) in recent years has been observed drastically. MRSA has become the deadliest gram-positive human pathogen that causes nosocomial infection (Mahmood et al., 2010). The key determinant of the wide range β-lactam resistance in MRSA strains is the Penicillin-Binding Protein 2a (PBP2a) (Lim & Strynadka, 2002). According to the pioneering fluid mosaic model of the cell, the various lipid species are divided into microdomains required in lipid raft formation which is necessary for cell membrane organization, a platform for protein-protein interactions control and cellular signal transduction (Wisniewska et al., 2003). Functional membrane microdomains were organized in a prokaryotic lipid-rafts region. *mecA* gene in MRSA builds resistance and does not allow antibiotics to bind transpeptidase enzyme involved in the cell wall of MRSA but allow it to normal replication (Hiramatsu et al., 1992). The gene encodes penicillin binding protein 2a (PBP2a) which has a low affinity towards β-lactam antibiotics as the active site of PBP2a is located inside the pocket and not accessible to β –lactams (Prasad et al., 2019). Currently, the drugs of choice for the treatment of infections by MRSA are vancomycin (1), daptomycin (2), ceftaroline (3) and linezolid (4). Even with controlled use, strains that show reduced susceptibility, or outright resistance, to these drugs have arisen (Appelbaum, 2006; Hiramatsu et al., 1997; Stryjewski & Corey, 2014; Turos et al., 2007; Vancomycin-Resistant Staphylococcus aureus Investigative Team, 2003). Furthermore, the clinical usefulness of vancomycin is expected to diminish because of the emergence of vancomycin-resistant *S. aureus* (VRSA) and vancomycin-intermediate *S. aureus* (VISA) (Appelbaum, 2006; Deresinski, 2007) (Figure 1).

There is an emergence of the new lead molecules to treat MRSA infections (Raghunath et al., 2012). Over the last few decades, even though researchers have attempted to address the rise of antimicrobial resistance by *in-silico* approaches there is an urgent need for new agents to treat patients with severe bacterial infections caused by antibiotic resistant strains (Anitha et al., 2016; Moreillon, 2008; Ragunathan et al., 2019). It is well known that *N*-heterocycles exhibited promising antibacterial activity (Radini, 2018). Recent research studies addressed the inhibition of the lipid A biosynthesis by the pyrimidine, pyridine and pyrazole analogues (Hu et al., 2019). Thus, this work mainly focuses on pyrimidine and pyridine, a class of *N*-heterocyclic compounds that have

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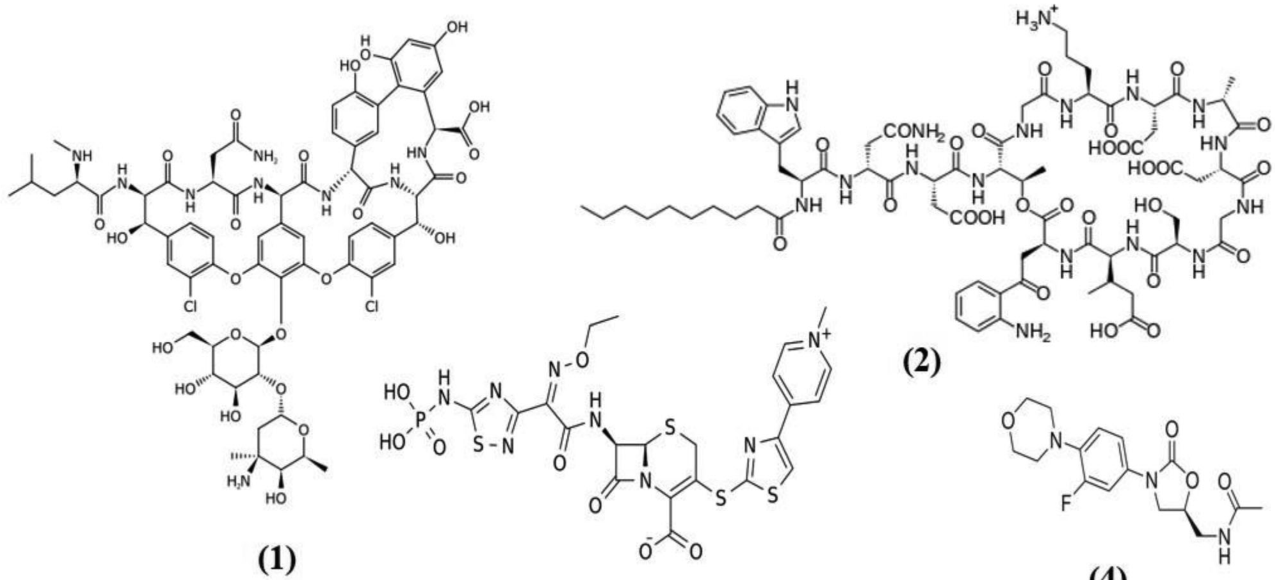
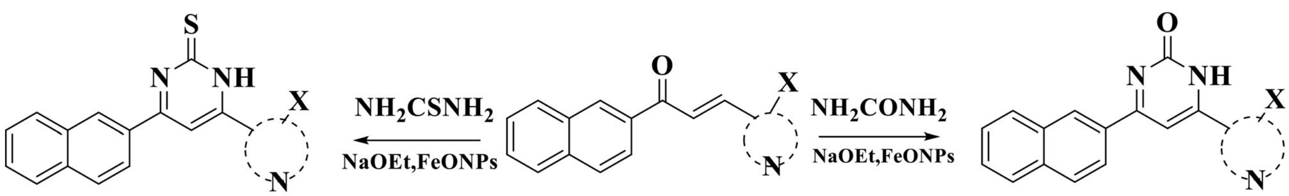


(1) (2) (3) (4)

Figure 1. Structure of drugs to treat MRSA infections.



5(a-g) 3(a-g) 4(a-g)

Scheme 1. Schematic representation for synthesis of Pyridine coupled Pyrimidinone/Pyrimidinthione.

been widely explored essentially because of the different medicinal and therapeutic properties (Prachayasittikul et al., 2017). The pyrimidinone skeleton is an essential part of numerous natural or synthetic bioactive materials (Madhavan et al., 2002). Although the design of small molecules with antibacterial activity is an attractive approach to fight against bacterial infections, the synthesis of this kind of pyrimidinone moiety is consistently an extraordinary challenge (Balalaie et al., 2008). In view of the mentioned significant characteristics, the present investigation designed to directly overcome MRSA infection. Thus, the new pyridine coupled pyrimidinones and pyrimidinthiones were designed, synthesized and characterized by different spectral techniques. The synthesized compounds were evaluated for the minimum inhibitory concentration (MIC), molecular docking studies and also molecular dynamics simulation (MDS) for the potent compounds.

2. Materials and method

Chemicals and reagents were procured from Sigma Aldrich (Merck Pvt. Ltd., India). For column chromatography, Silica gel was purchased from Merck Pvt. Ltd., (Mumbai, India). The completion of the reaction was monitored by TLC. The FT-IR spectra were recorded on a PerkinElmer Spectrum Version 10.03.09. The ^{1}H and ^{13}C spectra (Agilent-NMR) were recorded using $CDCl_3$/DMSO-d_6 as solvents. Chemicals shifts were reported in parts per million relatives to tetramethylsilane as an internal standard, Waters micro TOF QII mass spectrometer was used to determine the Mass.

2.1. General procedure for the synthesis of pyrimidinones/pyrimidinthiones derivatives

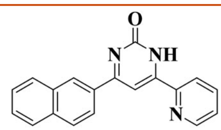
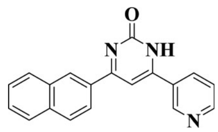
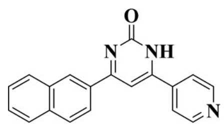
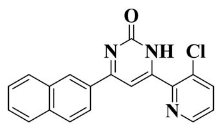
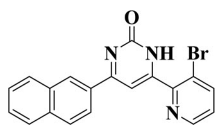
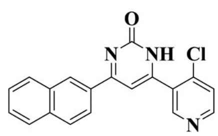
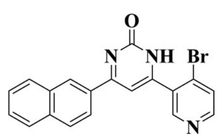
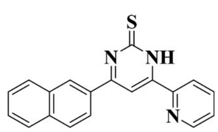
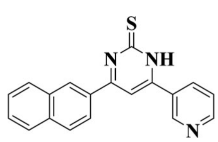
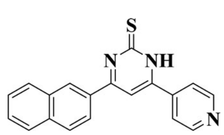
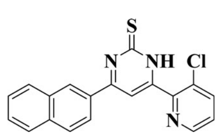
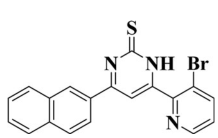
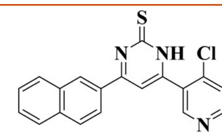
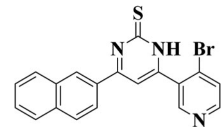
The equimolar concentration of pyridine chalcones 3(a-g) and urea or thiourea were dissolved in ethanol (15 mL). Sodium ethoxide and Iron oxide nanoparticles were added to the reaction mixture and were stirred at reflux conditions for 3-4 h. The reaction progress was monitored by thin-layer chromatography. The reaction mixture was filtered, and the filtrate was poured into crushed ice, the separated solids were filtered and recrystallized from ethanol to get the product 4(a-g) and 5(a-g) in better yields (Scheme 1).

2.2. Antibacterial studies

2.2.1. Minimum inhibitory concentration (MIC)

The minimum inhibitory concentration (MIC) of the test sample against the MRSA ATCC 43300 strain were determined using the broth microdilution method described by the CLSI standard (CLSI,2012). Each well of a 96-well microtiter plate was filled with 100 µL Mueller–Hinton broth (MHB). The test samples were dissolved in 10% dimethyl sulfoxide (DMSO) with a graded

Table 1. The structure and MIC values of the synthesized compounds.

Compound	Structure	MIC μg
4a		23
4b		28
4c		30
4d		25
4e		10
4f		20
4g		8
5a		45
5b		40
5c		42
5d		40
5e		52
5f		43
5g		55

(continued)

Table 1. Continued.

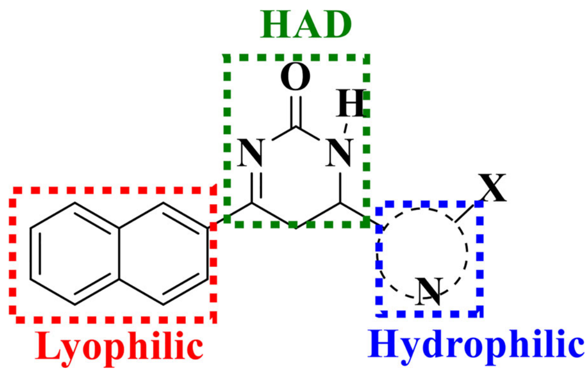


Figure 2. Pharmacophore model of the proposed title compound.

concentration in the range of (1-100 μg) was added to Column 1-10. The direct suspension inoculum was prepared by the turbidity of MacFarland standard 0.5 or approximately $1–2 \times 10^8$ CFU/ml and 40 μL of inoculum added to MHB. Subsequently, 30 μL of 0.2 μM Millipore membrane filtered 0.02% (wt/vol) resazurin was added to each test well and incubated at 37°C for 24 h. Column 11 served as a positive control medium with bacterial inoculums and column 12 served as a negative control with only medium. The standard antibiotic vancomycin was used. Any colour changes were observed. Blue/purple colour indicated no bacterial growth while pink/colourless indicated bacterial growth. The MIC value was taken at the lowest concentration of the test sample that inhibits the growth of bacteria (colour remained in blue).

2.3. Molecular docking

2.3.1. Protein preparation

To understand the interaction of ligand with protein, molecular docking studies were performed using AutoDock Vina 1.5.6 tool (Di Muzio et al., 2017). Three-dimensional structures of viral protein penicillin binding protein 2a (PBP2a) (PDB ID: 1MWT), was downloaded in PDB format from RCSB protein data bank website [https://www.rcsb.org/]. By using Biovia Discovery Studio 2019 visualizer in bound atoms were removed from the proteins (Biovia, 2017). The AutoDock tool were used to add the non-polar hydrogen atoms to the protein and was utilised to run and analyse the docking simulations. Finally, the structure of the receptor was saved in PDBQT format.

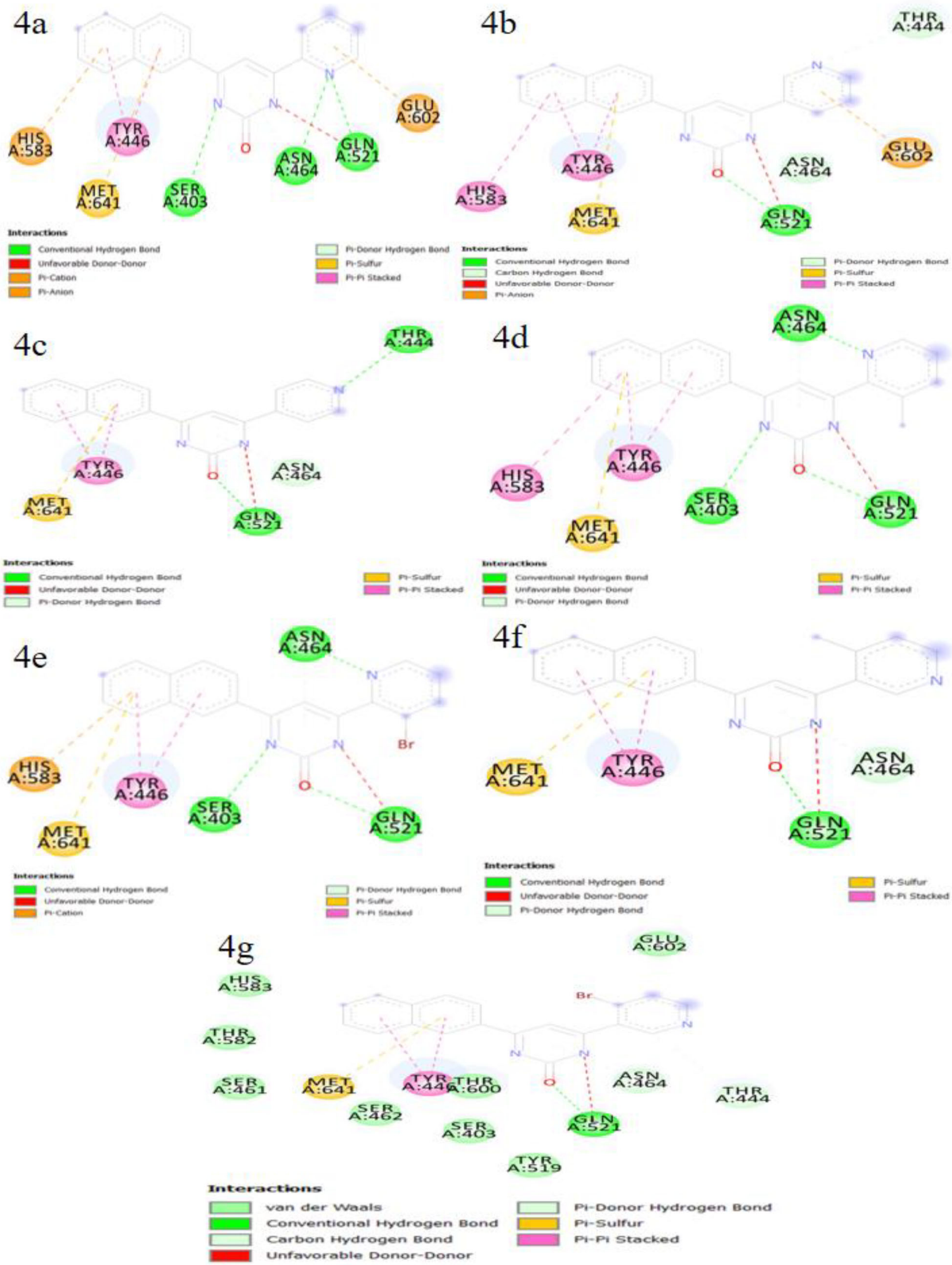


Figure 3. 2 D images of protein–ligand complex: Interaction between Protein PBP2a and synthesized compounds of 4(a-g).

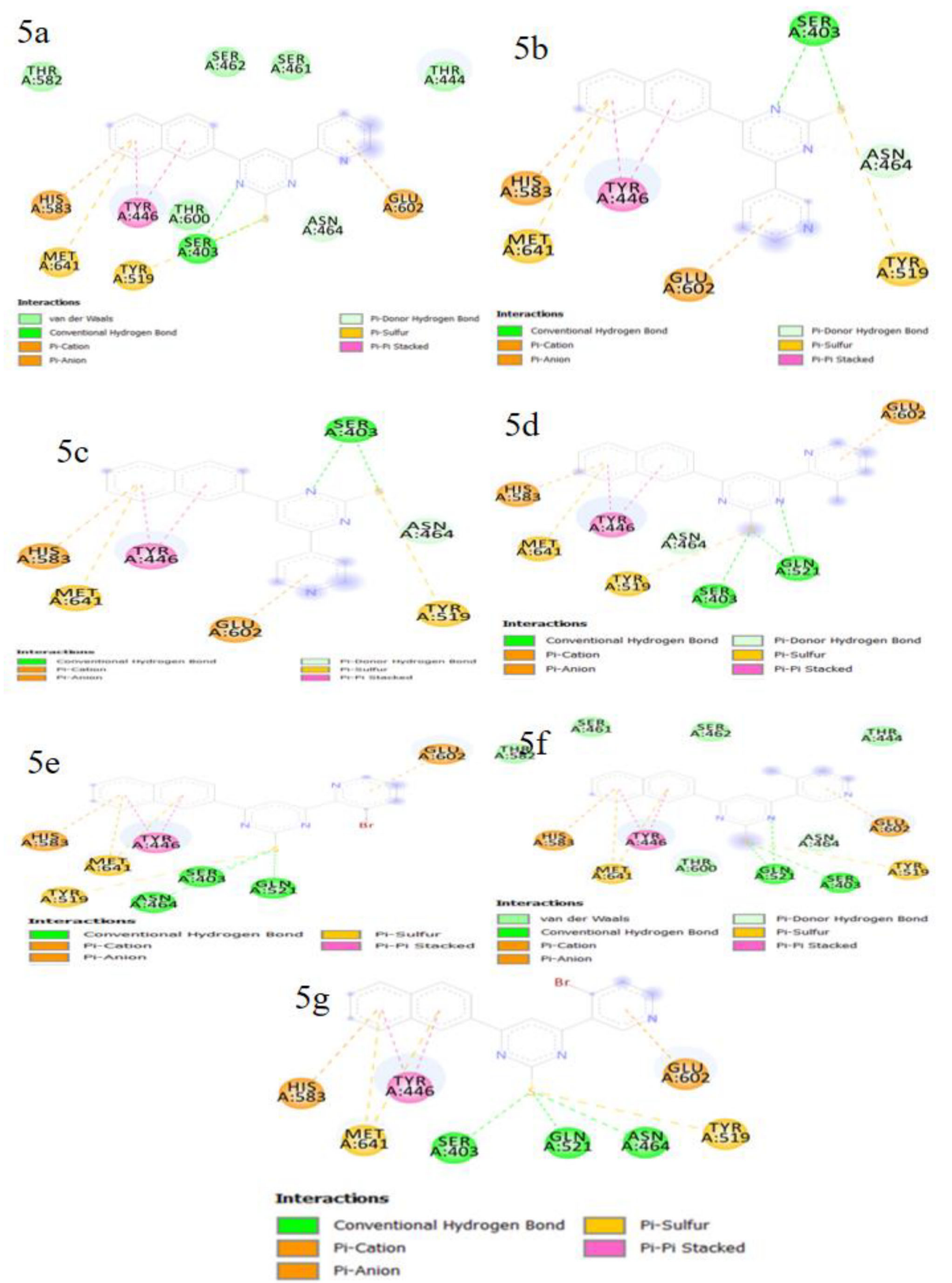


Figure 4. 2 D images of protein–ligand complex: Interaction between Protein PBP2a and synthesized compounds of 5(a-g).

Table 2. The docking score and the active site amino acids interactions with the synthesized compounds.

Compound	Residue involved	Ligand	Interaction type	Distance (Å)	Binding score (kcal/mol)
4a	GLN521	N of Pyridine	Hydrogen	2.22	−9.7
	GLN521	N of Pyrimidinone	Donor	2.45	
	ASN464	π of Pyrimidinone	Donor	2.74	
	SER403	N of Pyrimidinone	Hydrogen	2.89	
	ASN464	N of Pyridine	π- Sigma	3.05	
	GLU602	π of Pyridine	π-π stacked	3.87	
	TYR446	π of Naphthalene	π- Sigma	4.19	
	HIS583	π of Naphthalene	π Cation	4.93	
4 b	GLN521	O of Pyrimidinone	Hydrogen	2.40	−9.5
	GLN521	N of Pyrimidinone	Donor	2.64	
	ASN464	π of Pyrimidinone	π- Donor	2.65	
	GLU602	π of Pyridine	π-Anion	4.01	
	TYR446	π of Naphthalene	π-π stacked	4.10	
	TYR446	π of Naphthalene	π-π stacked	4.53	
4c	GLN521	O of Pyrimidinone	Hydrogen	2.44	−9.5
	ASN464	π of Pyrimidinone	π- Donor	2.60	
	GLN521	N of Pyrimidinone	Donor	2.69	
	THR444	N of Pyridine	Hydrogen	2.88	
	TYR446	π of Naphthalene	π-π stacked	4.16	
	TYR446	π of Naphthalene	π-π stacked	4.56	
	MET641	π of Naphthalene	π-Sulfur	5.02	
4d	ASN464	π of Pyrimidinone	π- Donor	2.60	−9.7
	GLN521	N of Pyrimidinone	Donor	2.65	
	GLN521	O of Pyrimidinone	Hydrogen	2.67	
	SER403	N of Pyrimidinone	Hydrogen	2.75	
	ASN464	N of Pyridine	Hydrogen	2.84	
	TYR446	π of Naphthalene	π-π stacked	4.44	
	TYR446	π of Naphthalene	π-π stacked	4.54	
4e	ASN464	π of Pyrimidinone	π- Donor	2.66	−9.7
	GLN521	N of Pyrimidinone	Donor	2.69	
	GLN521	O of Pyrimidinone	Hydrogen	2.74	
	SER403	N of Pyrimidinone	Hydrogen	2.75	
	ASN464	N of Pyridine	Hydrogen	2.85	
	TYR446	π of Naphthalene	π-π stacked	4.42	
	TYR446	π of Naphthalene	π-π stacked	4.49	
	HIS583	π of Naphthalene	π Cation	4.61	
4f	GLN521	O of Pyrimidinone	Hydrogen	2.38	−9.7
	SER195	π of Pyrimidinone	π- Donor	2.66	
	GLN521	N of Pyrimidinone	Donor	2.71	
	TYR446	π of Naphthalene	π-π stacked	4.13	
	TYR446	π of Naphthalene	π-π stacked	4.57	
	MET641	π of Naphthalene	π-Sulfur	5.03	
4g	GLN521	O of Pyrimidinone	Hydrogen	2.39	−9.8
	ASN464	π of Pyrimidinone	π- Donor	2.66	
	GLN521	N of Pyrimidinone	Donor	2.69	
	THR444	C of Pyridine	Carbon	3.62	
	TYR446	π of Naphthalene	π-π stacked	4.12	
	TYR446	π of Naphthalene	π-π stacked	4.57	
	MET641	π of Naphthalene	π-Sulfur	5.04	
5a	ASN464	π of Pyrimidinthione	Hydrogen	2.80	−9.0
	SER403	N of Pyrimidinthione	Hydrogen	2.95	
	SER403	S of Pyrimidinthione	Hydrogen	3.12	
	GLU602	π of Pyridine	π − Anion	3.69	
	TYR446	π of Naphthalene	π-π stacked	4.28	
	TYR446	π of Naphthalene	π-π stacked	4.46	
	HIS583	π of Naphthalene	π − Cation	4.90	
5b	ASN464	π of Pyrimidinthione	π- Donor	2.79	−9.1
	SER403	N of Pyrimidinthione	Hydrogen	2.94	
	SER403	S of Pyrimidinthione	Hydrogen	3.13	
	GLU602	π of Pyridine	π − Anion	3.67	
	TYR446	π of Naphthalene	π-π stacked	4.27	
	TYR446	π of Naphthalene	π-π stacked	4.46	
	HIS583	π of Naphthalene	π − Cation	4.91	
5c	ASN464	π of Pyrimidinthione	π- Donor	2.84	−9.1
	SER403	N of Pyrimidinthione	Hydrogen	2.95	
	SER403	S of Pyrimidinthione	Hydrogen	3.16	
	GLU602	π of Pyridine	π − Anion	3.55	
	TYR446	π of Naphthalene	π-π stacked	4.26	
	TYR446	π of Naphthalene	π-π stacked	4.46	
	HIS583	π of Naphthalene	π − Cation	4.92	
5d	GLN521	S of Pyrimidinthione	Hydrogen	2.26	−9.0
	GLN521	N of Pyrimidinthione	Hydrogen	3.07	
	ASN464	π of Pyrimidinthione	π- Donor	3.09	
	SER403	S of Pyrimidinthione	Hydrogen	3.35	

(*continued*)

Table 2. Continued.

Compound	Residue involved	Ligand	Interaction type	Distance (Å)	Binding score (kcal/mol)
	GLU602	π of Pyridine	π – Anion	3.60	
	TYR446	π of Naphthalene	π-π stacked	4.36	
	TYR446	π of Naphthalene	π-π stacked	4.43	
	HIS583	π of Naphthalene	π – Cation	4.70	
5e	GLN521	S of Pyrimidinthione	Hydrogen	2.18	−8.9
	SER403	S of Pyrimidinthione	Hydrogen	3.37	
	ASN464	S of Pyrimidinthione	Hydrogen	3.66	
	GLU602	π of Pyridine	π – Anion	3.67	
	TYR446	π of Naphthalene	π-π stacked	4.31	
	TYR446	π of Naphthalene	π-π stacked	4.42	
	HIS583	π of Naphthalene	π – Cation	4.67	
5f	GLN521	S of Pyrimidinthione	Hydrogen	2.21	−9.0
	GLN521	S of Pyrimidinthione	Hydrogen	3.05	
	ASN464	π of Pyrimidinthione	π- Donor	3.07	
	SER403	S of Pyrimidinthione	Hydrogen	3.37	
	GLU602	π of Pyridine	π – Anion	3.64	
	TYR446	π of Naphthalene	π-π stacked	4.36	
	TYR446	π of Naphthalene	π-π stacked	4.44	
	HIS583	π of Naphthalene	π – Cation	4.68	
5g	GLN521	S of Pyrimidinthione	Hydrogen	2.16	−8.9
	SER403	S of Pyrimidinthione	Hydrogen	3.41	
	ASN464	S of Pyrimidinthione	Hydrogen	3.67	
	GLU602	π of Pyridine	π – Anion	3.67	
	TYR446	π of Naphthalene	π-π stacked	4.30	
	TYR446	π of Naphthalene	π-π stacked	4.40	
	HIS583	π of Naphthalene	π – Cation	4.68	
Standard drug (Vancomycin)	GLN502	Hydrogen	Hydrogen	1.84	−7.8
	TYR344	Oxygen	Hydrogen	2.03	
	ASN505	Oxygen	Hydrogen	2.11	
	GLU602	Hydrogen	Hydrogen	2.37	
	GLY522	Hydrogen	Hydrogen	2.41	
	GLN521	Hydrogen	Hydrogen	2.69	
	TYR441	Oxygen	Hydrogen	2.73	
	GLN502	Hydrogen	Hydrogen	2.88	

2.3.2. Ligand preparation

The structure of ligands was drawn in chemdraw and converted into PDB format using Marvin JS software all the compounds were optimized and converted PDB to PDBQT format by using Auto Dock.

2.3.3. Grid box generation and virtual molecular docking

By using an Autodock vina software, binding energies were reported and determined in kcal/mol unit for ligand-protein. The grid box was generated by targeting the active site with a size of ($x=40$, $y=40$, $z=40$) and a centre of ($x=27.98$, $y=28.93$, $z=87.52$) was set to cover the binding site for PBP2a protein. Virtual molecular docking and analysis have been performed by using AutoDock Vina 1.5.6 tool. All the ligand molecules have been docked to active site of PBP2a protein. Finally, protein-ligand interactions were analysed by using Biovia discovery studio visualizer.

2.4. Molecular dynamics simulation

In order to examine the solidity of ligands in the binding pocket of protein and actions of protein complex during simulation, the molecular dynamic assay was carried. The docking complex of 1MWT protein was subjected to interaction with 4e and 4 g compound. To achieve the neutral condition of the protein complex system some quantity of Na^+ along with salt atoms was added. After upholding electro-neutralization, the complex was put through energy minimization on a convergence threshold of 1 kcal mol^{-1}Å by utilizing a conjugate gradient algorithm. Later on, on achieving energy minimization, the compound was bought into equilibrium using NPT gathering for 2 ns. The system was solvated using TIP3P water in a periodic box. The protein complexes had a minimum of 7 Å buffer in each course of the box to allow substantial fluctuations during the trail of MD simulation. The MD simulation analysis was carried out using Desmond modules of the Schrodinger 2020-2 suite (Fadaka et al., 2020; Sahoo et al., 2020).

2.4.1. Drug likeness and ADMET prediction

The standard pharmacokinetics parameters (ADMET) and drug-likeness were performed for the estimation of the physicochemical, drug-likeness and pharmacokinetics parameters in the drug discovery process. By using an Osiris datawarrior, the Physiochemical properties of the synthesized compounds were calculated and the parameters (Molecular weight, Clog P, Clog S, HBA, HBD, RB, TSA, and PSA) are represented in Table S1 in the supplementary file. The Swiss ADME online software was used for the prediction of ADMET (Adsorption, Distribution, Metabolism, Excretion and Toxicity) (Lucas et al., 2019). By using Swiss ADME drug design tool, the ADMET parameters and properties which include lipophilicity (iLogP, XLogP3, WLogP, MLogP, SILICOS-IT, $LogP_{o/w}$), water solubility-LogS (ESOL, Ali, SILICOS-IT), drug-likeness rules (Lipinski, Ghose, Veber, Egan and Muegge) and medicinal chemistry (PAINS, Brenk, Leadlikeness, Synthetic accessibility) methods are studied (Miller et al., 2020).

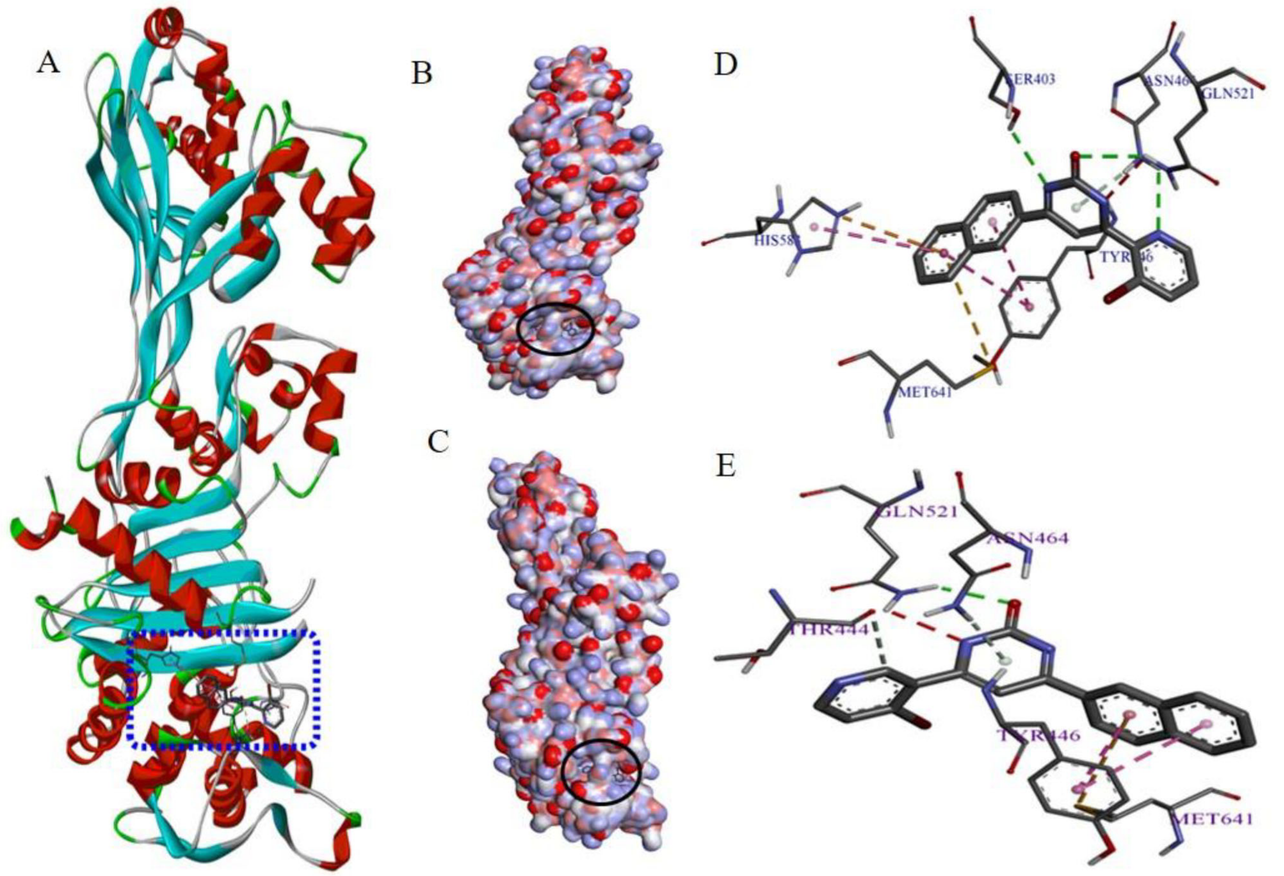


Figure 5. (A) PBP2a protein with active site (PDB: 1MWT); (B & C) Protein-ligand complex with 4e and 4 g in the active site; (D & E) Active site amino acids interactions with 4e and 4 g compounds.

3. Results and discussion

3.1. Chemistry

The pharmacophore model of the proposed structure is shown in Figure 2. The proposed structure is designed in such a way that the lyophilic unit is at one end and the hydrophilic unit is at another end. These two ends are linked with Hydrogen-Acceptor-Donor (HAD) unit. There are three units in the molecules,

- Naphthyl ring – Lyophilic unit
- Pyridine ring – Hydrophilic unit
- Pyrimidinone ring - HAD

The work involves the synthesis of pyrimidinones and pyrimidinthiones using pyridine chalcones with urea and thiourea. The structure of the obtained compounds has been characterized by IR, ^{1}H NMR and Mass spectra. The characterized data of synthesized compounds are given in the supplementary file. The IR spectra of pyrimidinone and pyrimidinthiones showed the presence of absorption bands in the region 1560-1650 cm^{-1} due to the presence of C$=$N. The absorption bands at 3200-3370 cm^{-1} confirmed the presence of the NH group. The ^{1}H NMR spectra revealed the presence of singlet at 8.8 ppm due to the presence of NH proton. The presence of multiplet in the region of 6.6-7.8 confirmed the presence of aromatic protons. The molecular ion peaks of the synthesized compounds were obtained with very good intensities in the recorded mass spectra.

3.2. Antibacterial assay

All the synthesised compounds were evaluated for *in-vitro* antibacterial activity on MRSA strain. The antibacterial data revealed, that the synthesized compounds showed very good antibacterial activity against MRSA strain and which is compared with the standard drug (Vancomycin) which showed 1 μg. The MIC values of the synthesised compounds are shown in Table 1. Among the pyrimidinone 4(a-g) and pyrimidinthione 5(a-g) series, pyrimidinone 4(a-g) showed better anti-MRSA activity compared to pyrimidinthione 5(a-g). The substituent of pyrimidine to pyridine ring is very essential for the antibacterial activity, thus the substituent at the different position was altered to study the effect of anti-MRSA activity. Among the substituents, 2-pyridine substituent (4a: 23 μg) showed better MIC value compared to 3- and 4-pyridine substituent (4 b: 28 μg & 4c: 30 μg) in pyrimidinone derivatives. The presence of Chloro and Bromo moiety enhanced the anti-MRSA activity in the designed compounds. Among the Chloro and Bromo compounds, Bromo showed good activity compared to chloro compounds. To understand the importance of Bromo moiety in the designed compounds, the positions of Bromo in the pyridine was altered. Among the Bromo substituent compounds, 4 g (8 μg)

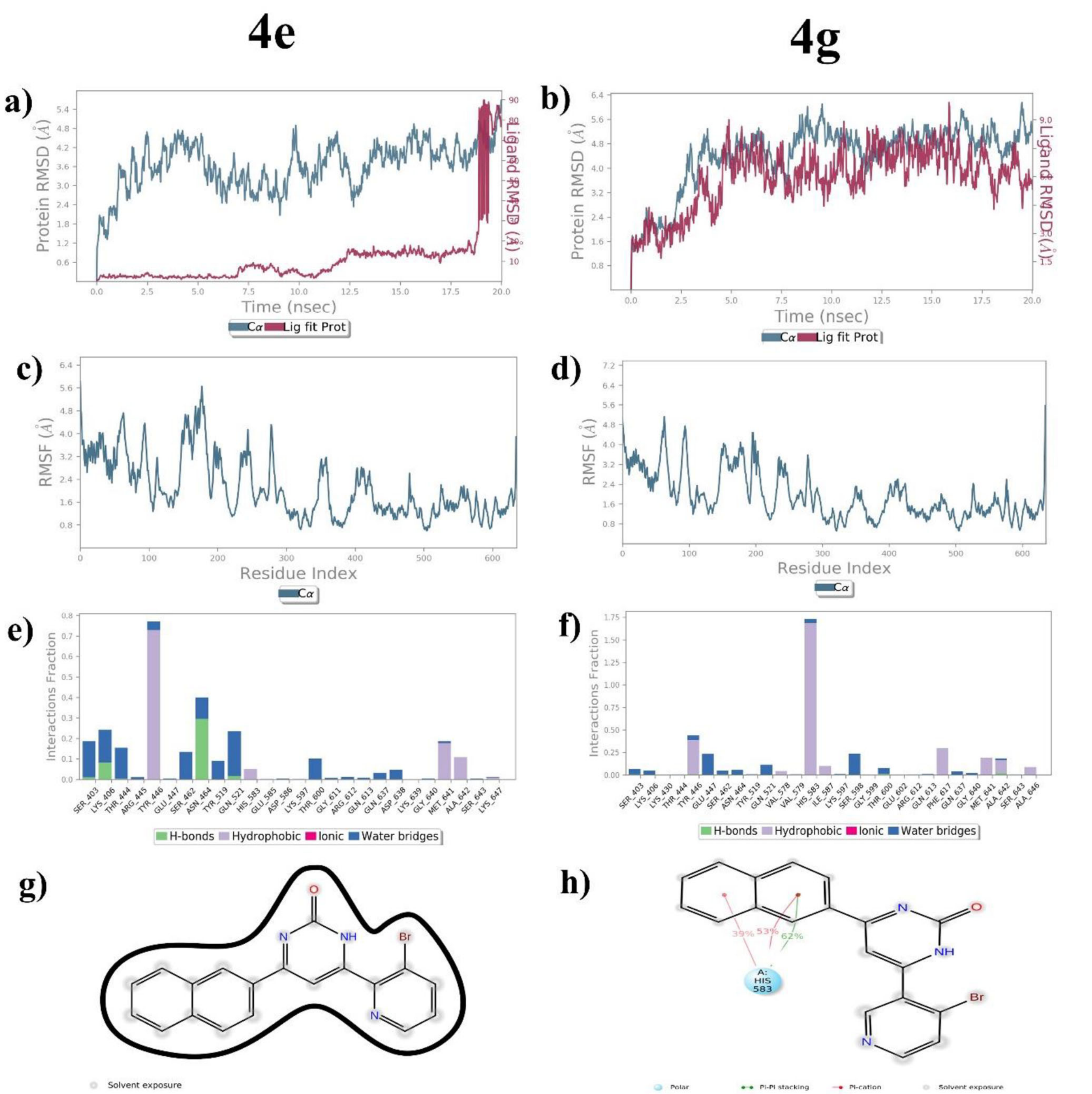


Figure 6. Molecular dynamic stimulation results of 4e and 4 g complexes. a) and b) RMSD analysis of 2 complexes indicating the interaction stability, c) and d) RMSF analysis of 2 complexes e) and f) signifies the number of H-bonds present between ligand and the protein molecule g) and h) indicating the interaction complex with amino acids of the protein molecule.

showed a very good antibacterial activity compared to 4e (10 µg). These synthesized compounds were compared with the standard drug (Vancomycin) which showed 1 µg.

3.3. Docking study

In order to support the experimental data which were obtained, the synthesized compounds were subjected to the docking studies. The docking study revealed that pyrimidinone 4(a-g) showed a better docking score compared to pyrimidinthione 5(a-g) derivatives. All the designed compounds showed good interaction with the amino acids in the active site of the PBP2a protein. The 2 D interaction images of synthesized compounds 4(a-g) and 5(a-g) with PBP2a protein are shown in Figures 3 and 4. The docking score and the active site amino acids interactions with the synthesized compounds are depicted in the Table 2. The docking study revealed various interactions such as H-bonding, π-sigma, π-donor, π-cation, hydrogen, π-alkyl, π-π-T shaped, etc of the synthesized compounds with different amino acids in the active site of the protein. The compound 4 g and standard drug showed conventional hydrogen bonding with amino acid residue GLN 521 with the docking score −9.8 and −7.8 Kcal/mol respectively. The interaction of vancomycin with PBP2a were incorporated in the Table 2 and Figure S_1 and Figure 5.

Table 3. Prediction of absorption in gastrointestinal of synthesized compounds.

	Absorption							
Compound	Water solubility (log mol/L)	Caco2 permeability (log Papp in 10^{-6} cm/s)	Intestinal absorption (human) (% Absorbed)	Skin Permeability (log Kp)	P-glycoprotein substrate	P-glycoprotein I inhibitor	P-glycoprotein II inhibitor	Water solubility (log mol/L)
4a	−3.357	1.321	96.35	−2.737	Yes	No	Yes	−3.357
4b	−3.453	1.293	95.778	−2.735	Yes	No	Yes	−3.453
4c	−3.458	1.284	95.993	−2.735	Yes	No	Yes	−3.458
4d	−3.454	1.36	95.048	−2.736	Yes	Yes	Yes	−3.454
4e	−3.472	1.365	94.981	−2.736	Yes	Yes	Yes	−3.472
4f	−3.517	1.335	95.134	−2.736	Yes	Yes	Yes	−3.517
4g	−.535	1.34	95.067	−2.736	Yes	Yes	Yes	-.535
5a	−3.504	1.475	93.615	−2.736	Yes	No	yes	−3.504
5b	−3.561	1.355	93.043	−2.735	Yes	No	Yes	−3.561
5c	−3.568	1.347	93.258	−2.735	Yes	No	Yes	−3.568
5d	−3.762	1.467	92.066	−2.736	Yes	No	Yes	−3.762
5e	−3.779	1.464	91.999	−2.736	Yes	No	Yes	−3.779
5f	−3.843	1.24	92.152	−2.736	Yes	No	Yes	−3.843
5g	−3.859	1.238	92.085	−2.736	Yes	No	Yes	−3.859

Table 4. Prediction of distribution in human body of synthesized compounds.

	Distribution			
Compound	VDss (human) (log L/kg)	Fraction unbound (human) (Fu)	BBB permeability (log BB)	CNS permeability (log PS)
4a	0.151	0.2	0.675	−1.892
4b	−0.051	0.185	0.359	−1.88
4c	−0.053	0.188	0.359	−1.879
4d	−0.052	0.151	0.468	−1.73
4e	−0.037	0.15	0.467	−1.708
4f	−0.133	0.148	0.449	−1.727
4g	−0.118	0.147	0.447	−1.704
5a	0.31	0.192	0.875	−0.949
5b	0.1	0.178	0.559	−0.738
5c	0.1	0.18	0.559	−0.753
5d	0.133	0.144	0.621	−0.908
5e	0.15	0.143	0.62	−0.908
5f	0.051	0.139	0.601	−0.859
5g	0.068	0.138	0.6	−0.859

Table 5. Prediction of metabolism and excretion in human body of synthesized compounds.

	Metabolism							Excretion	
Compound	CYP2D6 substrate	CYP3A4 substrate	CYP1A2 inhibitor	CYP2C19 inhibitor	CYP2C9 inhibitor	CYP2D6 inhibitor	CYP3A4 inhibitor	Total Clearance (log ml/min/kg)	Renal OCT2 substrate
4a	No	Yes	Yes	Yes	No	No	Yes	0.781	No
4b	No	Yes	Yes	Yes	No	No	No	0.871	No
4c	No	Yes	Yes	Yes	No	No	No	0.862	No
4d	No	Yes	Yes	Yes	No	No	Yes	0.152	No
4e	No	Yes	Yes	Yes	No	No	Yes	−0.09	No
4f	No	Yes	Yes	Yes	No	No	Yes	0.201	No
4g	No	Yes	Yes	Yes	No	No	Yes	−0.041	No
5a	No	Yes	Yes	Yes	No	No	Yes	0.203	No
5b	Yes	Yes	Yes	Yes	No	No	No	0.121	No
5c	Yes	Yes	Yes	Yes	No	No	No	0.043	No
5d	Yes	Yes	Yes	Yes	Yes	No	Yes	0.187	No
5e	Yes	Yes	Yes	Yes	Yes	No	Yes	−0.055	No
5f	Yes	Yes	Yes	Yes	No	No	Yes	0.236	No
5g	Yes	Yes	Yes	Yes	Yes	No	Yes	−0.005	No

3.4. Molecular dynamics simulation

Molecular docking scores sometimes do not reflect the ligand affinity (Hollingsworth & Dror, 2018) and inhibition in subsequent *in vitro* experiments. Thus, the *in-vitro* anti-MRSA activity and molecular docking study was validated with molecular dynamic simulations (MDS). Furthermore, the amino acid residues in the protein active site which are involved in the interaction with a ligand can be determined based on the resulting MDS confirmations. MDS investigate the intrinsic stability and conformational flexibility of the complexes (Srivastava et al., 2020). Utilizing the data from the MD simulation, RMSD, RMSF, the interactions were computed as an element of time. RMSD results showed the structural stability of the protein–ligand complexes within a systematic time (Yadav et al., 2018). RMSD plot and compound 4 g complex exhibited more stability throughout the simulation than 4e complex which is shown in Figure 6a and b. The event trajectory of the 4 g complex stayed stable through the MD simulation, with the accepted standards of

Table 6. Prediction of Toxicity in human body of synthesized compounds.

	Toxicity									
Compound	AMES toxicity	Max. tolerated dose (human) (log mg/kg/day)	hERG I inhibitor	hERG II inhibitor	Oral Rat Acute Toxicity (LD50) (mol/kg)	Oral Rat Chronic Toxicity (LOAEL) (log mg/kg_bw/day)	Hepatotoxicity	Skin Sensitisation	*T.Pyriformis* toxicity (log ug/L)	Minnow toxicity (log mM)
4a	Yes	0.572	No	Yes	3.486	1.105	Yes	No	0.288	1.84
4b	Yes	0.681	No	Yes	3.369	0.867	Yes	No	0.287	1.566
4c	Yes	0.676	No	Yes	3.343	0.818	Yes	No	0.288	1.598
4d	No	0.747	No	Yes	3.504	0.113	Yes	No	0.287	−0.892
4e	No	0.748	No	Yes	3.504	0.096	Yes	No	0.287	−1.038
4f	Yes	0.785	No	Yes	3.431	0.646	Yes	No	0.288	−1.14
4g	Yes	0.786	No	Yes	3.431	0.6299	Yes	No	0.288	−1.286
5a	Yes	0.599	No	Yes	3.459	0.853	Yes	No	0.288	1.29
5b	Yes	0.708	No	Yes	3.341	0.615	Yes	No	0.288	1.016
5c	Yes	0.704	No	Yes	3.315	0.566	Yes	No	0.288	1.049
5d	Yes	0.788	No	Yes	3.454	−0.003	Yes	No	0.289	−1.005
5e	No	0.789	No	Yes	3.454	−0.019	Yes	No	0.289	−1.151
5f	Yes	0.839	No	Yes	3.353	0.343	Yes	No	0.289	−1.253
5g	No	0.84	No	Yes	3.352	0.327	Yes	No	0.289	−1.399

change within 2.4-5.6 Å whereas, the event trajectory of the 4e complex is not stable. By RMSF analysis, the values of protein residues of the 4e and 4 g complexes showed that the initial terminal region within 100 residue index was less stable due to larger fluctuations in both the complexes. Whereas, the other residue region of both complexes did not exhibit much fluctuation indicating the stability of the complexes. The bonding and breaking of H bonds is an important component in MD simulation. The greater the interaction of H bond presence between ligand and protein, the greater is the binding stability. The compound 4 g binds in the active binding pocket of the 1MWT protein amino acid HIS583 with greater interaction. It binds with 39% and 53% pi-cation with naphthalene ring whereas binds with 62% pi-pi stacking with that of the naphthalene ring as shown in Figure 6h. In case of 4e compound no interaction with amino acids has been observed indicating the least binding ability of the compound with the complex (Figure 6g).

3.5. Drug likeness and ADMET prediction

In the field of drug discovery, many tools are used to predict the ADMET of the molecules. The intestinal absorption of all the compounds was found to have good absorption. The skin permeability constant (log Kp) of all the compounds are found to be greater than −2.5 which indicates that all the compounds possess low skin permeability. Whether the compounds are likely to be a P-glycoprotien substrate, P-glycoprotien I inhibitor and P-glycoprotien II inhibitor are also indicated in the Table 3. The steady state volume of distribution (VDss) is the theoretical volume that the total dose of a drug would need to be uniformly distributed to give the same concentration as in blood plasma. VDss is considered low if log VDss < -0.15 and high if above log VDss > 0.45. All the compounds showed moderate VDss as mentioned in the Table 4. The brain is protected from exogenous compounds by the blood brain barrier. For a given compound, logBB > 0.3 is considered to readily cross the blood brain barriers, while molecules with logBB < -1 are poorly distributed to the brain. It is proved by the data obtained that all the compounds selected showed the property to readily cross the blood brain barriers. It is known that the compounds with a logPS > -2 are considered to penetrate the central nervous system (CNS), while those with logPS < -3 are unable to penetrate the CNS. From the distribution data, it is confirmed that all the compounds are considered to penetrate the CNS. Whether all the compounds are likely to the isomers of P450 inhibitors or they are metabolised by CYP2D6 and CYP3A4 are listed in the Table 5. It is confirmed in the excretion table that all the compounds are not likely to be an OCT2 substrate and also the total clearance of all the compounds are mentioned. The toxicity parameters are listed in Table 6.

4. Conclusion

In the present research work, we designed and synthesized biologically potent pyridine linked pyrimidinone and pyrimidinthione analogs against MRSA. All the compounds showed a very good antibacterial activity against MRSA, among which 4e and 4 g showed better antibacterial activity. *In silico* ADMET, molecular docking and MD simulation studies were carried out to validate the *in-vitro* method. The potent compounds, 4e and 4 g were subjected to molecular dynamics simulation to analyse the stability of the compounds with the targeted PBP2a protein. MD simulations revealed computational predictions were consistent with the experimental results. Compound 4 g exhibited stable interactions up to 20 ns. We conclude from, Docking and MD simulation analysis that both protein and compound 4 g are stable in the binding pocket throughout the simulation. The results obtained from our study will be useful for designing effective compounds with potential therapeutic effects against MRSA. On further improvisation of these compounds, their application in the area of the biomedical field can be enhanced.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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